Q1)

I. Q1: 15 MARKS

A lecture is 60 minutes long. You arrive at the beginning of the lecture. The process of recording attendance is completed by a certain time selected uniformly and randomly during the lecture. You leave soon after attendance is recorded. Calculate the expected value of the time you spend attending the lecture?

The time you leave the lecture is the time after start that attendance is recorded. Therefore, the time is a Uniform RV $(0, 60)$.

Thus the expected value is

$$\int_0^{60} x \frac{1}{60}\, dx = \frac{1}{2}\frac{1}{60}(3600-0)$$
$$= 30 \text{ mins.}$$

Q2

II. Q2: 20 MARKS

You attend a 60 minute lecture. During every minute of the lecture your attention is demanded by the lecture slide, your WhatsApp application, and your Facebook application. You always pay attention to the lecture slide during the first and the last minute. For every other minute, you pay attention to the lecture slide with probability 0.4 and to the WhatsApp application with probability 0.5. Calculate the expected number of minutes during which you pay attention to the lecture slides?

60 minutes! Think 60 trials.

The first and last trials have an outcome of "lecture slide".

Each of the other 58 trials have an outcome

lecture slide w.p. 0.4,
WhatsApp w.p. 0.5
FB w.p 0.1 $(1-0.4-0.5)$.

The event of interest for each of these 58 trials is "lecture slide". So effectively each trial is a Bernoulli RV that maps the outcome "lecture slide" to 1 and "Whatsapp or FB" to 0. The parameter of the Bernoulli RV is 0.4.

I therefore have a Binomial$(58, 0.4)$ RV (58 Bernoulli trials of the above kind)

Average number of "lecture slides" over these 58 is simply the expected value of the Binomial$(58, 0.4)$. This value is $(58)(0.4)$. The other two trials (minutes) are known to have lecture slides. ∴ Therefore, expected no. of minutes of lecture slides is $(58)(0.4)+2 = 25.2$

[Note that we assumed that the outcomes of the lecture minutes are independent of each other. In the absence of any information that suggests otherwise, feel free to assume independence.]

III. Q3: 25 MARKS

Number of assignments in a probability and statistics course offered over the duration of 1 semester are often well modeled by the Poisson random variable of rate 20 per semester. Suppose half the assignments are 40 minutes long and the rest are 20 minutes long. Calculate the expected value of the time students spend on assignments? [Hint: The PMF of the Poisson RV K is $P[K=k] = \frac{(\lambda T)^k}{k!}e^{-\lambda T}$, where λ is the rate and T is the duration of observation.]

Number of assignments is the RV K.

$$P[K=k] = \frac{(\lambda T)^k e^{-\lambda T}}{k!}$$

Let $g(K)$ be the time spent on assignments. $g(K)$, which is a function of the RV K is given by

$$g(K) = \left(\frac{K}{2}\right)20 + \frac{K}{2}(40)$$
$$= \left(\frac{K}{2}\right)(60) = 30K.$$

We want

$$E[g(K)]$$
$$= \sum_{k=0}^{\infty} 30k\, P[K=k]$$
$$= 30 \sum_{k=0}^{\infty} k \frac{(\lambda T)^k e^{-\lambda T}}{k!}$$

For our problem, $\lambda = 20$ per semester
$T = 1$ semester.

$$\therefore E[g(K)] = 30 \sum_{k=0}^{\infty} k \frac{(20)^k e^{-20}}{k!}$$
$$= 30(\lambda T) = (30)(20) \text{ minutes}$$
$$= 600 \text{ minutes.}$$
$$= 10 \text{ hours.}$$

Q4

A bus may arrive with equal probability at the start of one of the 60 minutes (indexed $\{1, 2, \ldots, 60\}$) in an hour. You catch the bus in case you arrive at the same minute as the bus or you arrive earlier than the bus.

You arrive at the bus stop at minute 4. Calculate the probability that you miss the bus (arrived after the bus). Calculate the average value of the minute at the start of which the bus arrives.

After you arrive at minute 4, you are told that you have missed the bus. What is the average number of minutes by which you missed the bus?

P[You arrive at 4 and miss the bus]
= P[Bus arrived at minute 1 or 2 or 3]
= P[Bus arrived at 1] + P[Bus arrived at 2] + P[Bus arrived at 3]

$$= \frac{3}{60} = \frac{1}{20}.$$

Average value of the minute when the bus arrives is simply the expected value of the discrete Unif RV over $\{1, 2, \ldots, 60\}$

It is $$\sum (1+2+\cdots+60)\frac{1}{60}$$
$$= \frac{(60)(60+1)}{2}\frac{1}{60}$$
$$= \frac{61}{2} = 30.5 \text{ minutes.}$$

You arrive at minute 4 and are told that the bus has already left!.

Let the RV X denote the minute the bus arrives.

You are interested in

$$E[4-X \mid \text{Bus arrived before minute } 4]$$
$$= 4 - E[X \mid \text{Bus arrived before minute } 4]$$

$$P[X=x] = \begin{cases} \frac{1}{60}, & x \in \{1, 2, \ldots, 60\} \\ 0, & \text{otherwise} \end{cases}$$

$$P[X=x \mid \text{Bus arrived before minute } 4]$$
$$= \frac{P[X=x, \text{Bus arrived before } 4]}{P[\text{Bus arrived before } 4]}$$
$$= \begin{cases} \frac{P[X=x]}{P[\text{Bus arrived before } 4]} & x \in \{1,2,3\} \\ 0 & \text{otherwise} \end{cases}$$
$$= \begin{cases} \frac{1/60}{3/60} = \frac{1}{3} & x \in \{1,2,3\} \\ 0 & \text{otherwise} \end{cases}$$

∴ The expectation of interest is

$$4 - \left(1\left(\tfrac{1}{3}\right) + 2\left(\tfrac{1}{3}\right) + 3\left(\tfrac{1}{3}\right)\right)$$
$$= 4 - \left(\frac{6}{3}\right) = 2 \text{ minutes.}$$